Data-Driven Optimization

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과제 3

문제 1. 정수 모형 (계속)

PROTOTYPE EXAMPLE

The CALIFORNIA MANUFACTURING COMPANY is considering expansion by building a new factory in either Los Angeles or San Francisco, or perhaps even in both cities. It also is considering building at most one new warehouse, but the choice of location is restricted to a city where a new factory is being built. The *net present value* (total profitability considering the time value of money) of each of these alternatives is shown in the fourth column of Table 12.1. The rightmost column gives the capital required (already included in the net present value) for the respective investments, where the total capital available is \$10 million. The objective is to find the feasible combination of alternatives that maximizes the total net present value.

TABLE 12.1 Data for the California Manufacturing Co. example

Decision Number	Yes-or-No Question	Decision Variable	Net Present Value	Capital Required
1	Build factory in Los Angeles?	x_1	\$9 million	\$6 million
2	Build factory in San Francisco?	x_2	\$5 million	\$3 million
3	Build warehouse in Los Angeles?	x_3	\$6 million	\$5 million
4	Build warehouse in San Francisco?	x_4	\$4 million	\$2 million
			Capital available:	\$10 million

좌측 문제를 모델링하면 다음과 같다.

$$\text{Maximize} \quad Z = 9x_1 + 5x_2 + 6x_3 + 4x_4,$$

subject to

$$\begin{aligned}
6x_1 + 3x_2 + 5x_3 + 2x_4 &\le 10 \\
x_3 + x_4 &\le 1 \\
-x_1 + x_3 &\le 0 \\
-x_2 + x_4 &\le 0 \\
x_j &\le 1 \\
x_j &\ge 0
\end{aligned}$$

$$x_j \text{ is integer,} \quad \text{for } j = 1, 2, 3, 4.$$

$$x_j \text{ is binary,} \quad \text{for } j = 1, 2, 3, 4.$$

문제 1. 정수 모형

12.1-1. Reconsider the California Manufacturing Co. example presented in Sec. 12.1. The mayor of San Diego now has contacted the company's president to try to persuade him to build a factory and perhaps a warehouse in that city. With the tax incentives being offered the company, the president's staff estimates that the net present value of building a factory in San Diego would be $7 million and the amount of capital required to do this would be $4 million. The net present value of building a warehouse there would be $5 million and the capital required would be $3 million. (This option would be considered only if a factory also is being built there.)

The company president now wants the previous OR study revised to incorporate these new alternatives into the overall problem. The objective still is to find the feasible combination of investments that maximizes the total net present value, given that the amount of capital available for these investments is $10 million.

원 문제의 모델을 고려하여 좌측 문제를 모델링하고 해를 구하기 위한 파이썬 프로그램을 작성하여라.

문제 1. 정수 모형 - Model

```
Maximize
 Obj: +9 x[0] +5 x[1] +7 x[2] +6 x[3] +4 x[4] +5 x[5]
Subject to
 const_0: +6 x[0] +3 x[1] +4 x[2] +5 x[3] +2 x[4] +3 x[5]  <= 10
 const_1: +1 x[3] +1 x[4] +1 x[5]  <= 1
 const_2: -1 x[0] +1 x[3]  <= 0
 const_3: -1 x[1] +1 x[4]  <= 0
 const_4: -1 x[2] +1 x[5]  <= 0
Binaries
 x[0],  x[1],  x[2],  x[3],  x[4],  x[5]
End
```

```
In [43]: runcell(0, 'C:/Users/1007/Dr
OPTIMAL
목적함수값 =  17.0
x[0]  =  0.0
x[1]  =  1.0
x[2]  =  1.0
x[3]  =  0.0
x[4]  =  0.0
x[5]  =  1.0
```

번호	질문	의사결정변수	순 현재 가치	투자금액
0	공장 in LA?	X[0]	9	6
1	공장 in SF?	X[1]	5	3
2	공장 in SD	X[2]	7	4
3	창고 in LA	X[3]	6	5
4	창고 in SF	X[4]	4	2
5	창고 in SD	X[5]	5	3

문제 2. 정수 모형

Violating Proportionality.

The SUPERSUDS CORPORATION is developing its marketing plans for next year's new products. For three of these products, the decision has been made to purchase a total of five TV spots for commercials on national television networks. The problem we will focus on is how to allocate the five spots to these three products, with a maximum of three spots (and a minimum of zero) for each product.

Table 12.3 shows the estimated impact of allocating zero, one, two, or three spots to each product. This impact is measured in terms of the *profit* (in units of millions of dollars) from the *additional sales* that would result from the spots, considering also the cost of producing the commercial and purchasing the spots. The objective is to allocate five spots to the products so as to maximize the total profit.

TABLE 12.3 Data for Example 2 (the Supersuds Corp. problem)

Number of TV Spots	Profit: Product 1	Profit: Product 2	Profit: Product 3
0	0	0	0
1	1	0	−1
2	3	2	2
3	3	3	4

좌측 문제를 모델링하고 해를 구하기 위한 파이썬 프로그램을 작성하여라.

문제 2. 정수 모형 - Model

Maximize

Obj: +1 X00 -1 X02 +3 X10 +2 X11 +2 X12 +3 X20 +3 X21 +4 X22

Subject to

spots_capa_2: +1 X00 +1 X01 +1 X02 +2 X10 +2 X11 +2 X12 +3 X20 +3 X21 +3 X22 <= 5

each_product_0: +1 X00 +1 X10 +1 X20 <= 3

each_product_1: +1 X01 +1 X11 +1 X21 <= 3

each_product_2: +1 X02 +1 X12 +1 X22 <= 3

Binaries

X00 X01 X02 X10 X11 X12 X20 X21 X22

End

```
Objective value = 7.0
X00  = 0.0
X01  = 0.0
X02  = 0.0
X10  = 1.0
X11  = 0.0
X12  = 0.0
X20  = 0.0
X21  = 0.0
X22  = 1.0
```

TABLE 12.3 Data for Example 2 (the Supersuds Corp. problem)

	Profit		
	Product		
Number of TV Spots	1	2	3
0	0	0	0
1	1	0	−1
2	3	2	2
3	3	3	4

문제 3. 용량제약 입지선정 문제

5개 창고로 부터 6개 도시로 휘발유를 공급한다고 한다. 각 창고의 공급량, 각 도시의 수요량, 각 창고로 부터 수요로 톤당 수송가격과 창고를 사용할 때 고정비용이 아래 표와 같다. 이때 모든 도시의 수요을 만족하면서 전체 비용을 최소화하는 정수모형을 만들고 이에 대한 해를 구하는 프로그램을 작성하시오.

창고	수요 도시						월공급량(톤)	월 고정비용
	1	2	3	4	5	6		
A	1675	400	685	1630	1160	2800	18	7650
B	1460	1940	970	100	495	1200	24	3500
C	1925	2400	1425	500	950	800	27	3500
D	380	1355	543	1045	665	2321	22	4100
E	922	1646	700	508	311	1797	31	2200
월간수요(톤)	10	8	12	6	7	11		

문제 3. 용량제약 입지선정 문제 - Model

Minimize

Obj: +1675 X00 +400 X01 +685 X02 +1630 X03 +1160 X04 +2800 X05 +1460 X10 +1940 X11 +970 X12 +100 X13 +495 X14 +1200 X15 +1925 X20 +2400 X21 +1425 X22 +500 X23 +950 X24 +800 X25 +380 X30 +1355 X31 +543 X32 +1045 X33 +665 X34 +2321 X35 +922 X40 +1646 X41 +700 X42 +508 X43 +311 X44 +1797 X45 +7650 Y[0] +3500 Y[1] +3500 Y[2] +4100 Y[3] +2200 Y[4]

Subject to

demands_0: +1 X00 +1 X10 +1 X20 +1 X30 +1 X40 >= 10

demands_1: +1 X01 +1 X11 +1 X21 +1 X31 +1 X41 >= 8

demands_2: +1 X02 +1 X12 +1 X22 +1 X32 +1 X42 >= 12

demands_3: +1 X03 +1 X13 +1 X23 +1 X33 +1 X43 >= 6

demands_4: +1 X04 +1 X14 +1 X24 +1 X34 +1 X44 >= 7

demands_5: +1 X05 +1 X15 +1 X25 +1 X35 +1 X45 >= 11

supplies_5: +1 X00 +1 X01 +1 X02 +1 X03 +1 X04 +1 X05 <= 18

supplies_5_1: +1 X10 +1 X11 +1 X12 +1 X13 +1 X14 +1 X15 <= 24

supplies_5_2: +1 X20 +1 X21 +1 X22 +1 X23 +1 X24 +1 X25 <= 27

supplies_5_3: +1 X30 +1 X31 +1 X32 +1 X33 +1 X34 +1 X35 <= 22

supplies_5_4: +1 X40 +1 X41 +1 X42 +1 X43 +1 X44 +1 X45 <= 31

supplies_5_5: +1 X00 +1 X01 +1 X02 +1 X03 +1 X04 +1 X05 -1e+14 Y[0] <= 0

supplies_5_6: +1 X10 +1 X11 +1 X12 +1 X13 +1 X14 +1 X15 -1e+14 Y[1] <= 0

supplies_5_7: +1 X20 +1 X21 +1 X22 +1 X23 +1 X24 +1 X25 -1e+14 Y[2] <= 0

supplies_5_8: +1 X30 +1 X31 +1 X32 +1 X33 +1 X34 +1 X35 -1e+14 Y[3] <= 0

supplies_5_9: +1 X40 +1 X41 +1 X42 +1 X43 +1 X44 +1 X45 -1e+14 Y[4] <= 0

All X[i, j] 는 정수, All Y[i] 는 이진변수

창고	수요 도시						월공급량 (톤)	월 고정비용
	1	2	3	4	5	6		
A	1675	400	685	1630	1160	2800	18	7650
B	1460	1940	970	100	495	1200	24	3500
C	1925	2400	1425	500	950	800	27	3500
D	380	1355	543	1045	665	2321	22	4100
E	922	1646	700	508	311	1797	31	2200
월간수요(톤)	10	8	12	6	7	11		

문제 3. 용량제약 입지선정 문제 – Code

```
from ortools.linear_solver import pywraplp

solver = pywraplp.Solver.CreateSolver("SCIP")

infinity = solver.infinity()

BIGM = 100000000000000

tcosts = [
    [1675, 400, 685, 1630, 1160, 2800],
    [1460, 1940, 970, 100, 495, 1200],
    [1925, 2400, 1425, 500, 950, 800],
    [380, 1355, 543, 1045, 665, 2321],
    [922, 1646, 700, 508, 311, 1797]
    ]
supplies = [18, 24, 27, 22, 31]
demands = [10, 8, 12, 6, 7, 11]
fixed_costs = [7650, 3500, 3500, 4100, 2200]

nCity = len(demands)
nWarehouse = len(supplies)
```

```
# 의사결정 변수
# i = 0, 1, 2(# of Spots), j = 0, 1, 2 (제품)
X = {}
for i in range(nWarehouse):
    for j in range(nCity):
        X[i, j] = solver.IntVar(0, infinity, "X"+str(i)+str(j))

# 각 창고에 대한 고정비 제약
Y = {}
for i in range(nWarehouse):
    Y[i] = solver.IntVar(0, 1, "Y[%i]" % i)
```

문제 3. 용량제약 입지선정 문제 – Code

```
# 제약조건
const_expr = []
# 수요 제약
for j in range(nCity):
    const_expr = [X[i, j] for i in range(nWarehouse)]
    solver.Add(sum(const_expr) >= demands[j], 'demands_'+str(j))

#  공급 제약
for i in range(nWarehouse):
    const_expr = [X[i, j] for j in range(nCity)]
    solver.Add(sum(const_expr) <= supplies[i], 'supplies_'+str(j))

#  고정비 제약
for i in range(nWarehouse):
    const_expr = [X[i, j] for j in range(nCity)]
    solver.Add(sum(const_expr) <= BIGM*Y[i], 'supplies_'+str(j))
```

```
# 목적함수
obj_expr = []
for i in range(nWarehouse):
    obj_expr.append(fixed_costs[i]*Y[i])
    for j in range(nCity):
        obj_expr.append(tcosts[i][j]*X[i, j])

solver.Minimize(solver.Sum(obj_expr))

# 모델 파일 생성
with open('hw3-2.lp', "w") as out_f:
    lp_text = solver.ExportModelAsLpFormat(False)
    out_f.write(lp_text)

status = solver.Solve()

if status == pywraplp.Solver.OPTIMAL:
    print("Objective value = %.1f" % solver.Objective().Value())
    for i in range(nWarehouse):
        for j in range(nCity):
            print(X[i, j].name(), " = %.1f" % X[i, j].solution_value())
    for i in range(nWarehouse):
        print(Y[i].name(), " = %.1f" % Y[i].solution_value())
else:
    print("The problem does not have an optimal solution.")
```

문제 3. 용량제약 입지선정 문제 – Code

```
Objective value = 44943.0
X00  = 0.0
X01  = 8.0
X02  = 0.0
X03  = 0.0
X04  = 0.0
X05  = 0.0
X10  = 0.0
X11  = 0.0
X12  = 0.0
X13  = 0.0
X14  = 0.0
X15  = 0.0
X20  = 0.0
X21  = 0.0
X22  = 0.0
X23  = 6.0
X24  = 0.0
X25  = 11.0
X30  = 10.0
X31  = 0.0
X32  = 12.0
X33  = 0.0
X34  = 0.0
X35  = 0.0
X40  = 0.0
X41  = 0.0
X42  = 0.0
X43  = 0.0
X44  = 7.0
X45  = 0.0
Y[0]  = 1.0
Y[1]  = 0.0
Y[2]  = 1.0
Y[3]  = 1.0
Y[4]  = 1.0
```